

2 Giant axonal neuropathy (GAN) is a rare autosomal recessive disease, which affects neurones.

(a) Mitochondria can move freely within the axons of neurones.

In GAN, the axons of neurones become enlarged and blocked by the accumulation of specific proteins. This prevents the free movement of mitochondria and affects the transmission of action potentials along the axon membrane.

(i) Suggest **and** explain why preventing the free movement of mitochondria within the cytoplasm of the axon affects the transmission of action potentials along the axon membrane.

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(ii) A nerve conduction velocity (NCV) test can be used to measure the speed of transmission of nerve impulses along neurones in different parts of the body.

NCV tests were carried out on three people with GAN. The speed of transmission of nerve impulses was measured in neurones in two nerves:

- the median nerve, one of the main nerves in the arm
- the peroneal nerve, one of the main nerves in the leg.

Table 2.1 shows the NCV test results for the three people with GAN and the expected range for a person without GAN.

Table 2.1

location of neurones	speed of transmission of nerve impulse/ ms^{-1}			
	person 1	person 2	person 3	expected range
median nerve	47	39	43	50–65
peroneal nerve	44	22	28	40–45

With reference to Table 2.1, describe the effect of GAN on the speed of transmission of nerve impulses.

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- (iii) Walking requires nervous control to coordinate movements. One of the first signs of GAN is having problems with walking.

Explain how the speed of transmission of nerve impulses in people with GAN can affect walking.

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(b) GAN is caused by a mutation in a gene that codes for a protein known as gigaxonin.

Scientists have tested gene therapy in mice with GAN. In one study, viral vectors containing a functioning allele of the gene that codes for gigaxonin were made. Mice with GAN were treated with **one** dose of these viral vectors at 12 months of age.

Six months after treatment, when the mice were 18 months old, the scientists used a rotarod test to measure the effect of the gene therapy.

In the rotarod test:

- the length of time the mice are able to balance on a moving platform is recorded
- the longer the length of time the mice can balance on the moving platform, the better their neurone activity.

The rotarod test was repeated on the same mice each month until the mice were 23 months old. The rotarod test was also carried out, at the same time intervals and ages, on mice with GAN that were **not** treated with gene therapy and on mice without GAN. All mice were kept in the same conditions.

The results are shown in Fig. 2.1.

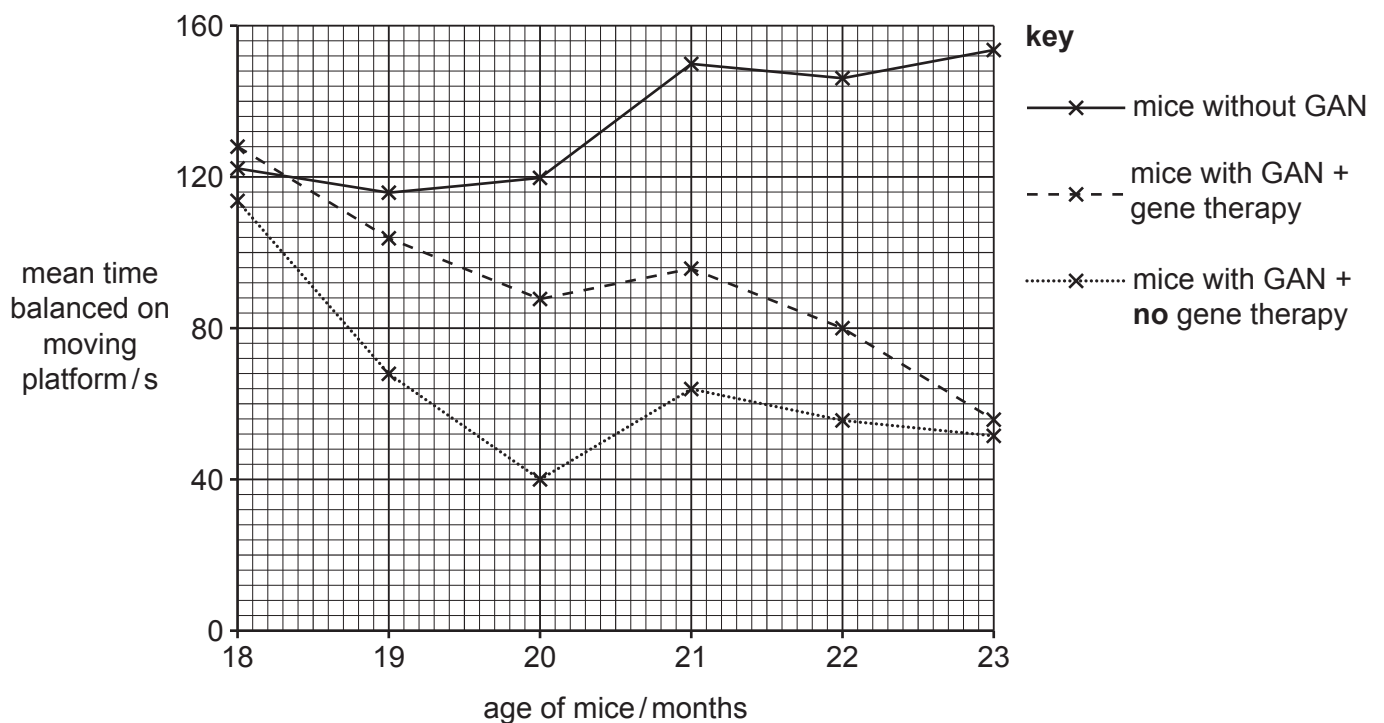


Fig. 2.1

From the results, some students concluded that:

- giving the mice with GAN one dose of gene therapy had a benefit, but did **not** cure the mice
- the results of the rotarod test were **not** affected by the age of the mice.

With reference to Fig. 2.1, discuss whether each of these **two** conclusions is justified by the data.

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Question 3 starts on page 9.